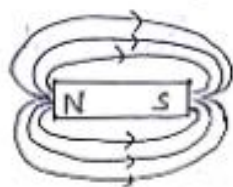


UNIT-3

TRANSFORMERS

Basic Definitions:-

Magnetic field:- The area which attracts magnetic materials and iron pieces, that region is called "magnetic field".



Magnetic flux (ϕ):- It is the total number of lines of force existing in a magnetic field is called as "magnetic flux (ϕ). Units are 'webers'."

Magnetic flux density: (B)

The flux per unit area in a magnetic field is called "magnetic flux density (B)".

i.e. $\boxed{B = \frac{\phi}{A}}$ $\rightarrow$ Units are "web/m²"

Magnetic field strength: (or) Magnetic field intensity (H)

It is the force experienced by unit North pole at particular point in a magnetic field is called "MFI".

Units are AT/web (or)

Ampere-Turns/web

1) Magneto motive force (mmf) :- The force which drives the flux through a magnetic material is called "mmf". units are "AT".

Permeability (μ) :- It is the ability of the medium which allows the magnetic fluxes through it self, is called permeability:

(or)
It is the ratio of flux density (B) to the magnetic field strength

$$\mu = B/H$$

$$(or) \boxed{\mu = \mu_0 \mu_r}$$

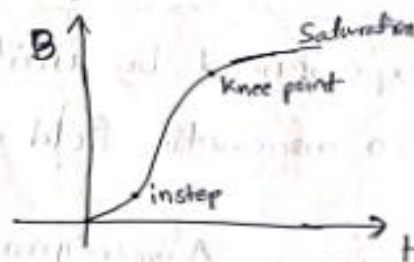
→ ~~unit~~ No unit

μ_0 = Absolute permeability
= 1 for air

μ_r = relative permeability

• B-H Curve of Magnetic material:-

→ The graph between flux density (B) and field intensity (H) is called B-H Curve.



Transformer :-

The transformer is a static device which transfers electrical power ^{from} one circuit to another circuit, at constant frequency.

- The transformer is power constant device
- The transformer is constant frequency device
- It can be used for different voltage & current changing values of A.C Supply
- It is used in different stages of electrical system as generation, transmission, distribution and utilization.

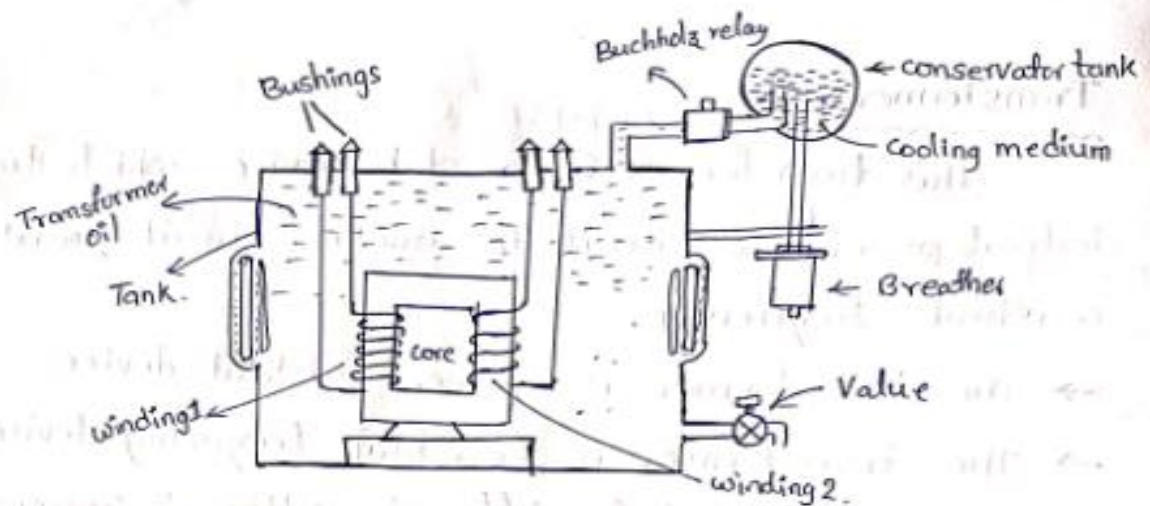
Construction of Transformer

The essential components of the transformer are

- ① Magnetic Core
- ② Two windings, namely primary and secondary windings
- ③ A time varying magnetic flux.

The constructional diagram of single-phase transformer as shown in below, it having different components as

- | | |
|----------------------|----------------------|
| ① Core | 6. Cooling medium |
| 2. Limb | 7. Breather |
| 3. Yoke or Tank | 8. Explosion vent |
| 4. Windings | 9. Buchholz relay |
| 5. Conservator Tank. | 10. Bushings. |
| | 11. Transformer oil. |



constructional diagram of Single-phase Transformer

① Transformer core:- The main function of transformer core which gives accommodation for windings transformer. It may be rectangular or square, having limbs. It is made up of cast iron. It is two types

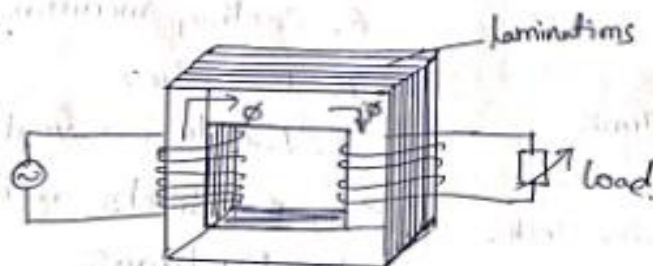
(a) core type

(b) shell type

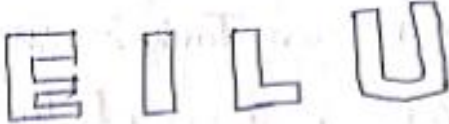
(a) core type transformer:- In this transformer rectangular frame laminations are formed to build the core.

→ It has single magnetic circuit

→ In this transformers windings are surrounded by limbs of core.



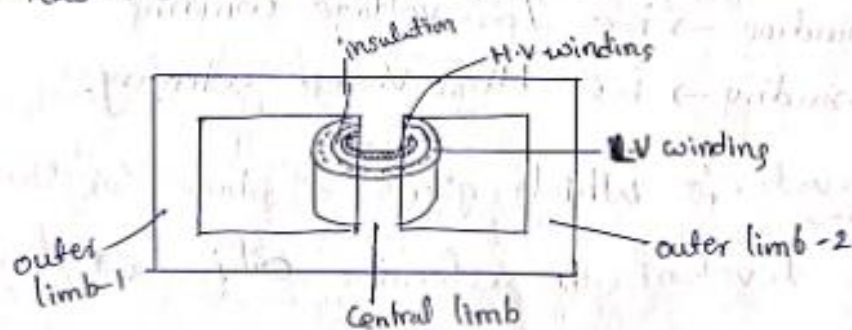
- The laminations are pressed or punched out from the steel sheets, we can ~~also~~ manufacture the core of Transformer
- These laminations sheets are assembled as letters

E, I, L, U as 

- In this type of core, interleaving windings is used which reduces leakage fluxes.

(b) Shell type transformer :- In this shell type windings are surrounded by core of transformer.

- It has double magnetic circuit.
- It has a central limb and two outer limbs, as



- For construction of this type core, we use 'E & I' shape lamination sheets.
- It requires less insulation for designing it.
- Cooling mechanism is easy.

2. Limb:- Limb is a part of core, which gives accommodation for winding and protect the winding for physical stress.

3. Yoke or Tank:- ~~It is a total cover~~

Yoke is part of transformer which protects transformer winding from physical stress and gives accommodation for transformer oil.

4. Windings:- We have two windings in transformer

→ Primary winding → which is connected with supply

→ Secondary winding → which is connected with load.

(or)

L.V winding → i.e Low voltage winding

H.V winding → i.e High Voltage winding.

5. conservator:- Which gives a place for increasing decreasing levels of transformer oil i.e a cooling medium

6. Breather:- Which^{is} protects the transformer oil from all weather condition problems.

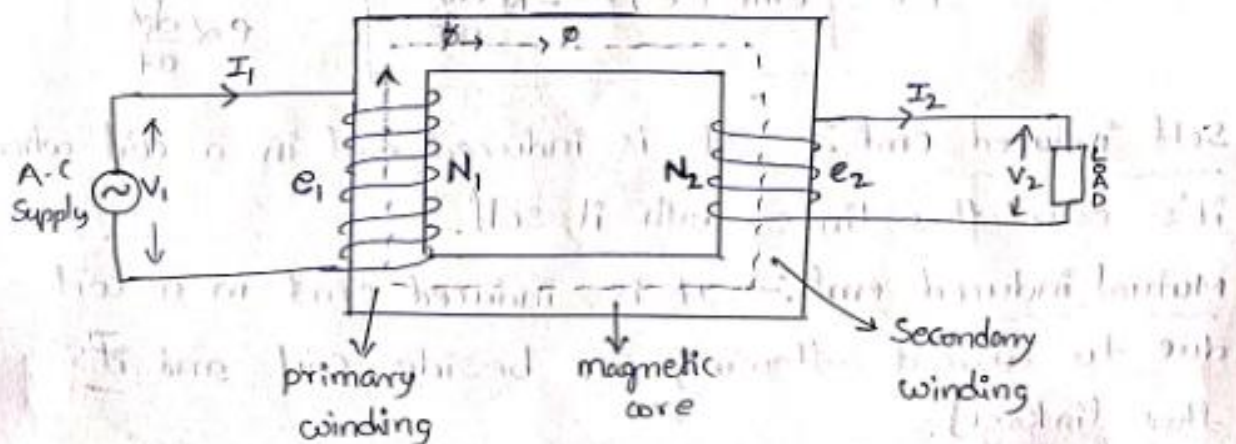
7. Buchholz relay:- The relay which rings the alarm when fault is occurred in transformer.

Working Principle of Transformer

The transformer is a static device which transfers electrical energy from one circuit to another circuit without change in frequency.

The transformer works on the principle of Faraday's Laws of electro-magnetic mutual induction principle, which states that "when two coils are magnetically coupled, then current in one winding changes uniformly, then that induces and develops the emf in other coil."

consider a basic transformer, which is having " N_1 " of turns in primary and " N_2 " no. of turns in secondary as shown in below.



- When primary winding is connected with AC source then current ' I_1 ' flows through the winding and produces an alternating flux (ϕ) in primary winding.
- These flux induces self induced emf (e_1) and passes through transformer magnetic core and links with secondary winding.
- According to Faraday's law of electro-magnetic mutual induction principle, an emf is induced in the secondary winding is called "mutual induced emf (e_2)".
- This secondary induced emf (e_2) causes the current (I_2) when secondary is connected to the load.
- At this situation the total mmf balance equation of transformer is

$$I_1 N_1 = I_2 N_2$$

- Based above balance equation transformer transfers electrical power from one circuit to other circuit without any electrical connection.
- Transformer transfers the power only based on magnetic coupling of two windings.

EMF - equation of transformer :-

In a transformer "when the primary winding is excited by an alternating voltage " V_1 ", it circulates alternating current produces an alternating flux ' ϕ '.

The wave form of alternating flux ' ϕ ' can drawn as for time period " 2π ", as shown below.

Let

ϕ = alternating flux

ϕ_m = maximum flux

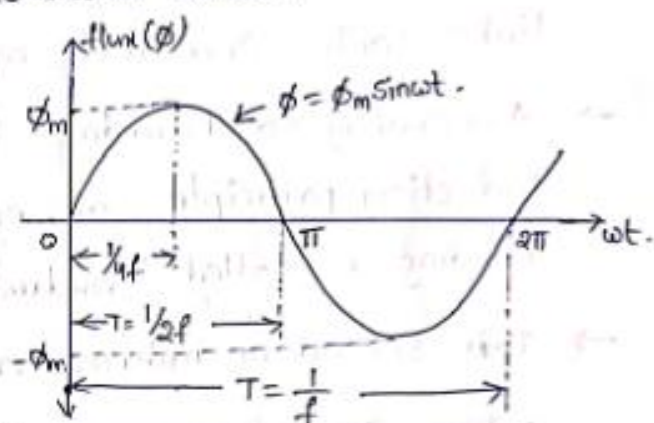
N_1 = No. of primary turns

N_2 = No. of secondary turns

E_1 = Primary induced emf in RMS

E_2 = Secondary induced emf in RMS

f = frequency of supply voltage.



→ According to Faradays law of electromagnetic induction the average emf induced in each turn is proportional to the rate of change of flux

i.e Average emf per turn = Average rate of change of flux

$$\text{i.e } \frac{d\phi}{dt} = \frac{\text{change in flux}}{\text{Time required change in flux}}$$

problems:- ① The maximum flux density in the core of 250/3000 volts, 50 Hz single phase transformer is 1.2 weber per square meter. If the emf per turns is 8 volts determine primary and secondary turns and area of the core.

P② The number of turns on the primary and secondary windings of a single phase transformer are 350 and 35 respectively. If the primary is connected to a 2.2 kV, 50 Hz, determine the secondary voltage.

Comparison between core type and shell-type transformer

Core type transformer

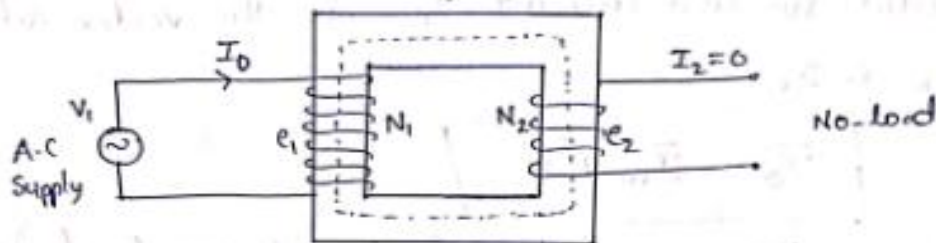
1. The windings surround the core and placed on the side limbs
2. Laminations is in L, I, or U shapes formats.
3. Two limbs exist
4. Requires more copper material
5. Requires more insulation
6. Equal distribution of flux on the limbs
7. More loss exists
8. High output ~~per~~ achieved
9. Easy to maintain
10. Natural cooling possible

Shell-type transformer

1. The core surrounds the winding and is placed on central limb.
2. Laminations sheet is in long strips in shape of E & I
3. Three limbs exist
4. Requires less copper
5. Less insulation require
6. Unequal flux distribution of flux in the limbs.
7. Less loss exists
8. Less output can be achieved
9. Difficult to maintain
10. Cooling mechanism exist

Practical Transformer on No-load

The practical transformer with no-load is shown in figure.



When transformer is on No-load, then secondary current is zero, but practical transformer has hysteresis and eddy current loss because it is excited by AC-source which causes alternating flux. Due to small internal resistance^{in winding} of transformer it ~~have~~^{causes} a small amount of primary copper loss. At this the current drawn by primary is called no-load current (I_0).

→ This no-load input current has two components

→ Wattless component : (I_m) : The magnetizing current (I_m) is the purely reactive component of I_0 , that lags V_1 by 90° , which produces the magnetic flux.

⊗ i.e. $I_m = I_0 \sin \phi_0$

→ Wattful component : (I_c) :- The power component (I_c) which supplies the total losses of transformer.

under no-load condition

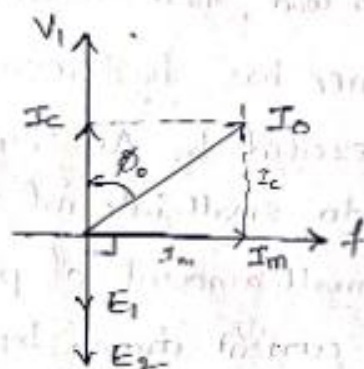
$$\text{i.e. } I_c = I_0 \cos \phi_0$$

This I_c is in phase with V_1 .

→ The total no-load current ' I_0 ' is the vector addition of I_m & I_c

$$\vec{I}_0 = \vec{I}_m + \vec{I}_c$$

→ The phasor diagram of transformer at no-load is



→ From the above phasor diagram, magnitude of no-load current

$$I_0 = \sqrt{I_m^2 + I_c^2} \quad \& \quad \cos \phi_0 \text{ is No-load power factor}$$

→ The total power input on no-load is

$$W_0 = V_1 I_0 \cos \phi_0 = V_1 I_c = P_i \text{ (Iron losses)}$$

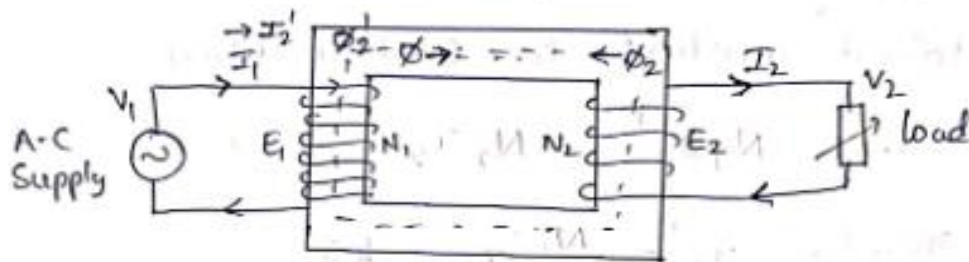
→ Where ' I_0 ' is very small, hence primary copper loss is negligible.

→ Hence power input ' W_0 ' on no-load always represent the iron losses as copper loss is negligible.

(8)

Transformer ON-load :-

The practical transformer connected with load is shown as below figure.



When primary winding of the transformer is connected to AC source, then V_1 induces emf E_2 in Secondary by transformer action. When secondary winding is connected with load then ' I_2 ' flows in winding.

- The current I_2 flowing through secondary winding produces flux " Φ_2 ", which opposes Φ .
- Then main flux Φ decreases, then E_1 decreases then current drawn by primary is " I_2' ", which produces additional flux ' Φ_2' ' which opposes Φ_2 .
- This I_2' neutralizes Φ_2 , the flux is maintained constant, hence transformer is flux constant machine.

→ The secondary flux ϕ_2 is develops the mmf is ' $N_2 I_2$ '
 and ϕ_2' flux produces mmf " $N_1 I_2'$ " these
 mmf are get balanced, the flux in transformer is
 maintained constant for load condition

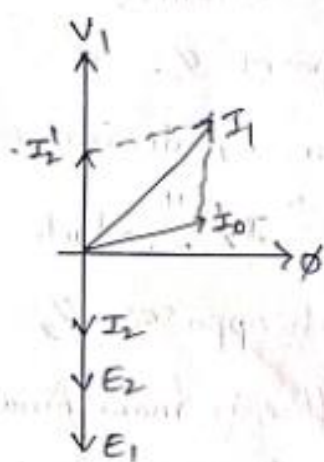
$$\text{i.e. } N_1 I_2' = N_2 I_2$$

$$\text{Therefore } I_2' = \frac{N_2}{N_1} I_2 = k I_2$$

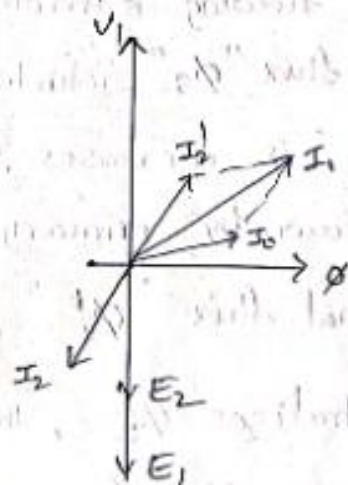
→ When a transformer is loaded, the primary current I_1
 has two components

$$\text{i.e. } \bar{I}_1 = \bar{I}_0 + \bar{I}_2'$$

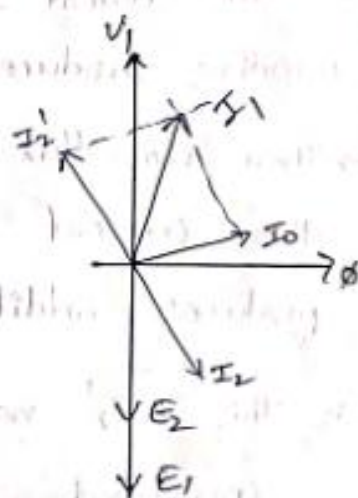
→ It is different for different loads, as in phasor diagram



Resistive-load



Inductive-load



Capacitive-load

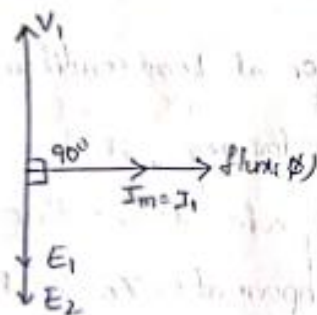
9

Ideal Transformer

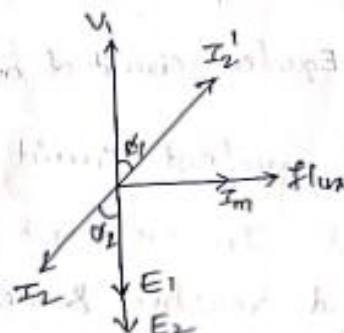
A transformer is said to be ideal transformer if it satisfies following properties

- It has "no losses"
- Its windings (i.e primary & secondary) have "zero resistance".
- Leakage flux is zero i.e 100% flux produced by primary links with the secondary.
- Permeability of core is so high.
- It has 100% efficiency.
- Magnetic coupling (K) = 1

Phasor diagrams for ideal transformer



Ideal transformer at No-load



Ideal transformer at Load.

In ideal transformer, voltage drops in primary and secondary are zero, because resistance is zero. and E_2 is equal to V_2 .

Equivalent Circuit of Transformer :-

The equivalent circuit of a transformer is a circuit with the combination of various resistances and reactances of the primary and secondary windings, which exactly perform like a transformer. It can be drawn as below.

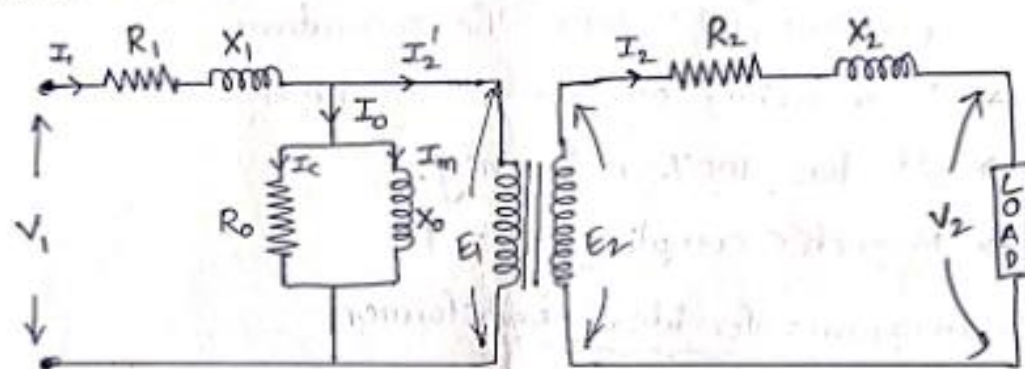


Fig:- Equivalent Circuit of a Transformer at Load condition

→ In equivalent circuit of transformer, it has no-load current I_0 are I_m & I_c which are flows through no-load reactive & active components X_0 & R_0

$$\text{where } R_0 = \frac{V_1}{I_c} \text{ and } X_0 = \frac{V_1}{I_m}$$

→ In equivalent circuit, of transformer has leakage reactances X_1 & X_2 w.r.to primary & secondary windings, which represent leakage flux

→ Winding resistance R_1 & R_2 to represent the copper losses $I_1^2 R_1$ & $I_2^2 R_2$ in the primary and secondary windings. respectively.

→ When load is connected, I_2 current flows and it causes voltage drop across R_2 & X_2 , then additional current $I_2' = I_2 k$.

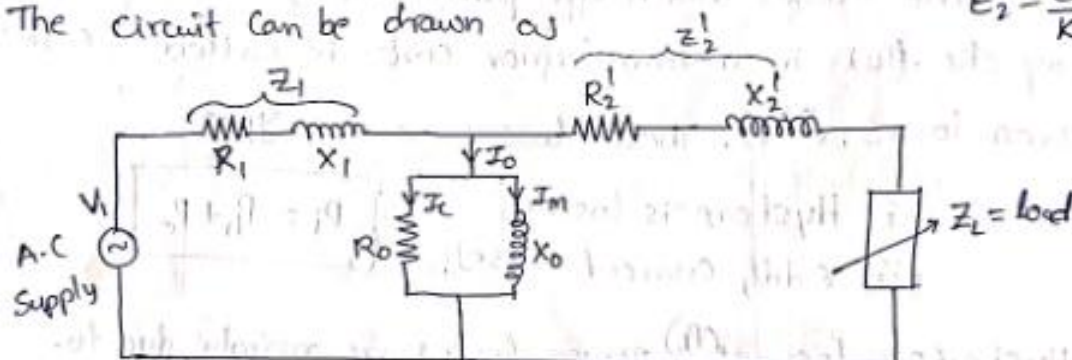
Equivalent circuit of transformer when referred to secondary-side-parameters referred to primary side

→ Then

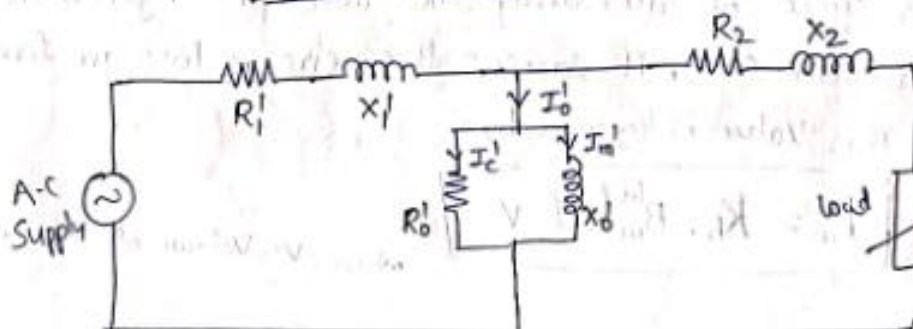
$$R_2' = \frac{R_2}{k^2}, \quad X_2' = \frac{X_2}{k^2}, \quad Z_2' = \frac{Z_2}{k^2} \quad \& \quad I_2' = k I_2$$

$$E_2' = \frac{E_2}{k}$$

The circuit can be drawn as



Equivalent circuit of Transformer when primary referred to secondary side:-



where

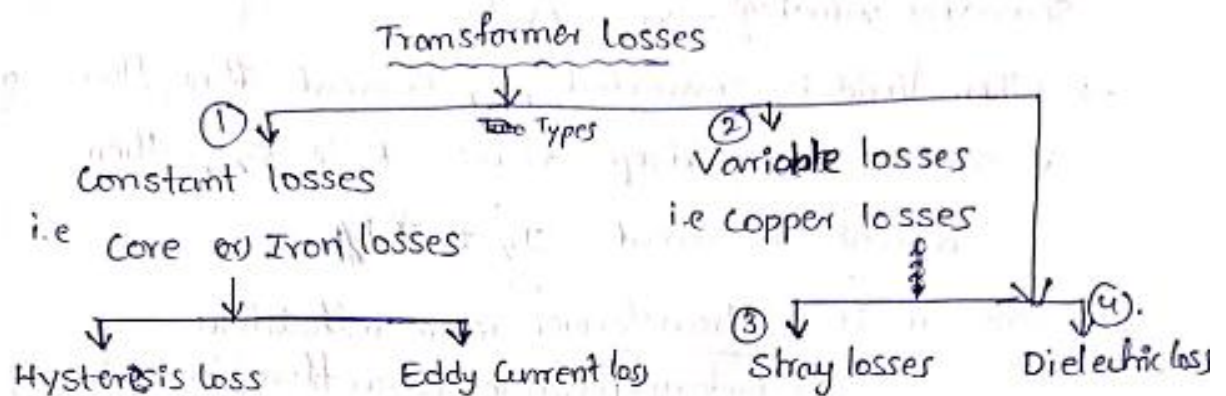
$$R_1' = k^2 R_1$$

$$X_1' = k^2 X_1$$

$$E_1' = k E_1$$

Losses in Transformer

→ Losses in transformer can be classified as



① Core (or) Iron losses (P_i)

The losses which are produced by alternating magnetic flux in a transformer core is called "Core or Iron losses (P_i)". These losses are two types

(i) Hysteresis losses

(ii) Eddy current losses.

$$P_i = P_h + P_e$$

(i) Hysteresis losses (P_h) These losses are mainly due to alternating flux in transformer core when transformer core is magnetization and de-magnetization takes place. The each cycle of alternating flux develops hysteresis loop in transform core, it causes the energy loss in form of heat. This value is given by

$$P_h = K_h \cdot B_m^{1.67} \cdot f \cdot V$$

where V = Volume of the core.

(ii) Eddy current loss : (P_e)

An induced emf in the core causes the current called eddy current, this eddy current circulates in transformer core and ~~the~~ causes " I^2R " losses. The losses due to eddy current in transformer core is called Eddy current losses.

$$i.e. \quad P_e = K_e B_m^2 f^2 t^2 V$$

where t = thickness of sheets.

These constant losses can be reduced (or) minimised by using high-grade silicon steel lamination sheets.

(2) Copper losses (P_c)

The losses due to winding resistances of transformer are called as "copper losses". The total copper loss.

$$P_c = I_1^2 R_1 + I_2^2 R_2$$

(3) Stray losses :- The losses that occur in the transformer due to the leakage of magnetic flux are called stray losses. These losses are very less percentage, so there are negligible.

(4) Dielectric loss :- The losses due to the insulating material for transformer are called as "Dielectric loss". This loss can be affect the efficiency of transformer.

Efficiency of The transformer

The efficiency of the transformer can be defined as its the ratio of the output power to the input power

$$\text{i.e. } \eta = \frac{\text{output power}}{\text{Input power}} \quad \text{or} \quad \frac{\text{Power output}}{\text{Power Input}}$$

$$\text{or } \eta = \frac{\text{output power}}{\text{Output power} + \text{Total losses}}$$

$$\eta = \frac{\text{Power output}}{\text{Power output} + P_i + P_{cu}}$$

by basis power equation

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{e2}} \quad (1) \quad \because P_{cu} = I^2 R \quad (2)$$

If the 'VA' rating of the transformer is $V_2 I_2$ then

$$\eta = \frac{(\text{VA rating}) \cos \phi_2}{(\text{VA rating}) \cos \phi_2 + P_i + I_2^2 R_{e2}}$$

Let $n = \frac{\text{Actual load}}{\text{Full load}}$ i.e. fraction of load, then

$$\eta = \frac{n \times (\text{VA rating}) \cos \phi_2}{n (\text{VA rating}) \cos \phi_2 + P_i + n^2 P_{cu}}$$

$$\eta \% = \frac{n (\text{VA rating}) \cos \phi_2}{n (\text{VA rating}) \cos \phi_2 + P_i + n^2 P_{cu}} \times 100$$

Condition for maximum efficiency (η_{max})

In a transformer the output power is less than input power, because it has the losses.

i.e. Power output $P_o = \text{Power input} - \text{Losses}$

$$P_o = P_i - P_{loss}$$

$$\therefore P_{loss} = P_i + P_{cu}$$

$$P_i = P_o + (P_i + P_{cu})$$

→ The load current at which efficiency attains maximum value is denoted by ' I_{2max} ' and the maximum efficiency denoted as " η_{max} ".

→ The efficiency is a function of load i.e. load current I_2 assuming $\cos\phi_2$ is constant, and secondary voltage ' V_2 ' also assumed as constant.

→ So condition for maximum efficiency is

$$\frac{d\eta}{dI_2} = 0$$

$$\text{We have } \eta = \frac{V_2 I_2 \cos\phi_2}{V_2 I_2 \cos\phi_2 + P_i + I_2^2 R_2} \quad \therefore \frac{d}{dI_2} \left(\frac{u}{v} \right) = \left(\frac{u'v - uv'}{v^2} \right)$$

$$\text{So } = (V_2 I_2 \cos\phi_2 + P_i + I_2^2 R_2)(V_2 \cos\phi_2) - (V_2 I_2 \cos\phi_2)(V_2 \cos\phi_2 + 0 + 2I_2 R_2) = 0$$

Let common " $V_2 \cos\phi_2$ " and cancel it, then

$$\rightarrow V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_2 - V_2 I_2 \cos \phi_2 - 2 I_2^2 R_2 = 0$$

$$\Rightarrow P_i - I_2^2 R_2 = 0$$

$$\boxed{P_i = I_2^2 R_2} \quad \text{i.e. condition for maximum efficiency}$$

i.e.

$$\boxed{\text{Iron losses} = \text{Copper losses}}$$

$$\boxed{P_i = P_{cu}}$$

Voltage Regulation :-

Not necessary

The voltage regulation of transformer can be defined as change in the ratio of change in secondary voltage when the transformer load is reduced from full load to no-load to the full load secondary voltage

where if V_2 = Secondary voltage at full load
 V_{02} = Secondary voltage at no-load

$$\therefore \text{Voltage Regulation} = \frac{V_{02} - V_2}{V_2}$$

$$\boxed{\% \text{ Voltage Regulation} = \frac{V_{02} - V_2}{V_2} \times 100}$$

→ If we have $V_2 = I_2 X_{02} \sin \phi$

$$V_{02} = I_2 R_{02} \cos \phi$$

Then

$$\% V.R = \frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{V_2}$$

for
inductive load = +ve
capacitive load = -ve

$$= \frac{I_2 R_{02} \cos \phi}{V_2} \pm \frac{I_2 X_{02} \sin \phi}{V_2}$$

→ The Regulation will be zero when power factor

angle $\phi = \tan^{-1} \left(-\frac{R_{02}}{X_{02}} \right)$

where '-ve' indicates that zero regulation occurs at leading power factor.

→ Regulation will be maximum when angle

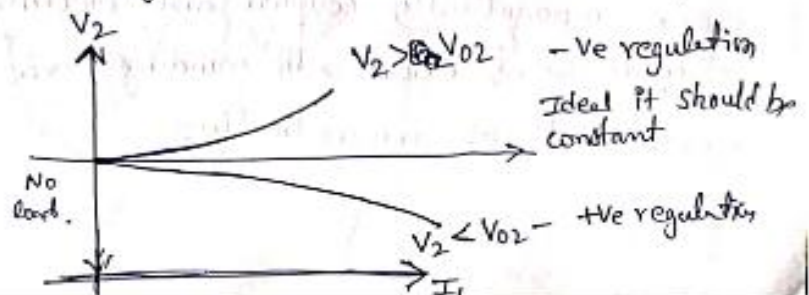
$$\phi = \tan^{-1} \left(\frac{X_{02}}{R_{02}} \right)$$

Maximum Regulation occurs at lagging power factor.

→ Voltage Regulation of transformer on an average is about 4%

→ For consumers view Regulation should be kept as possible

→ It can be shown as



Auto-transformer :-

An auto-transformer is a special type of single-phase transformer, consisting of single winding wound on a laminated core. Here, some part of this single winding acts as the primary winding and some part acts as the secondary winding.

- The number of turns of the primary and secondary windings can be varied using switch contact. Since the output voltage of the auto-transformer can be varied.
- So, it also called as VARIAC (VARIABLE A) (or) voltage regulator.

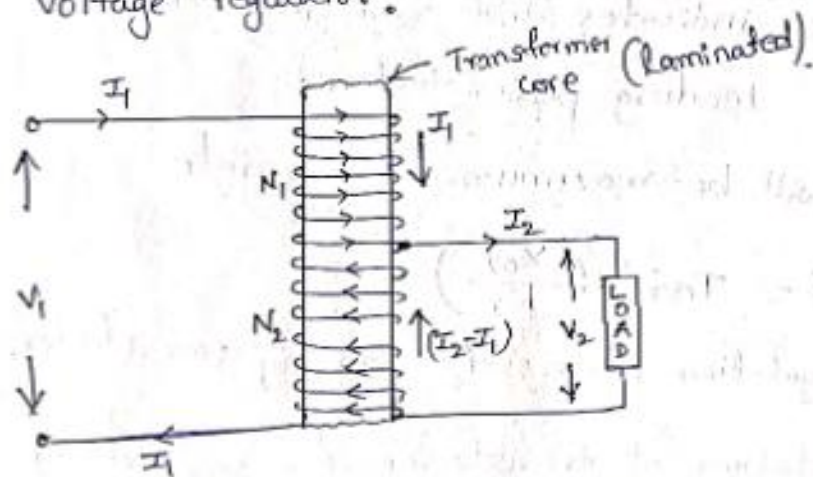


Diagram of Basic Auto-Transformer

- In two-winding transformer, the windings (primary & secondary) are magnetically coupled and electrically isolated, but in auto transformer both windings are connected electrically as well as magnetically.

iii)



When a supply voltage V_1 is applied to the primary winding AB, an alternating flux set up in the core due to which an induced emf ' E_1 ' is developed.

Since the secondary winding is a part of the primary winding, a part of the induced emf E_1 , is taken in to which the load is connected..

→ Let E_2 be the induced emf in the secondary winding.

→ This induced emf drives the current in the secondary winding and hence the load connected to it.

→ As in an ordinary transformer, the transformation ratio is turn ratio a

$$K = \frac{N_2}{N_1} = \frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

→ The power delivered to load is $P_L = V_2 I_2$

→ But in Auto transformer

$$P_{\text{Auto}} = V_2 I_2 (1-k)$$

→ & The ~~comp~~ power conducted directly is $P_c = K V_2 I_2$

→ When $N_1 > N_2$ → is Step-down Transformer

$N_1 < N_2$ → is Step-up Transformer.